Evaluation runs: chain (s4-chain)

generated 2026-11-05 from brain-data 0799b9755 2026-11-05 11:44:26 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): (the single pair): expected conforms (–threshold 65536 is needed at d¿=512). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
32	1	1	conforms	0.0	4	1 h, 250 GB	1	1	brain03, 2026-11-02
128	1	1	conforms	0.1	14	1 h, 250 GB	1	1	brain03, 2026-11-02
1024	1	1	conforms	2.1	407	48 h, 250 GB	1	1	brain01, 2026-11-03
2048	1	1	conforms	9.4	1471	48 h, 250 GB	1	1	brain04, 2026-11-03
4096	1	1	conforms	35.5	5545	48 h, 250 GB	1	1	brain04, 2026-11-03
8192	1	1	conforms	160.3	21491	48 h, 250 GB	1	1	brain04, 2026-11-03

Not run (stateless). left out by the table rules (ok): 512 fifo.
the tool ended with an error (exit-101): 16384 fifo.
planned but never scheduled (grid points beyond the ladder): 32768 fifo.

stateful

stateful: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8	1	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
32	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
128	1	1	conforms	0.1	12	1 h, 250 GB	brain02, 2026-11-02
1024	1	1	conforms	1.5	330	48 h, 250 GB	brain01, 2026-11-03
2048	1	1	conforms	5.3	1199	48 h, 250 GB	brain04, 2026-11-03
4096	1	1	conforms	21.1	4548	48 h, 250 GB	brain04, 2026-11-03
8192	1	1	conforms	108.9	14820	48 h, 250 GB	brain04, 2026-11-03
32768			🕓 (48 h)			48 h, 250 GB	brain01, 2026-11-03

Not run (stateful). left out by the table rules (ok): 512 fifo, 16384 fifo.

offline

offline: pair **single**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	1	conforms	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain01, 2026-11-02
8	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
32	1	1	conforms	0.0	4	1 h, 250 GB	0.0	0.0	0.0	merge	brain01, 2026-11-03
128	1	1	conforms	0.2	10	1 h, 250 GB	0.2	0.1	0.0	merge	brain01, 2026-11-03
512	1	1	conforms	0.5	80	48 h, 250 GB	0.2	0.4	0.1	merge	brain01, 2026-11-03
4096	1	1	conforms	28.5	3697	48 h, 250 GB	7.1	22.8	5.7	merge	brain01, 2026-11-03

Not run (offline). skipped: a smaller size had reached its limit: 1024 fifo, 16384 fifo, 32768 fifo.
the tool ended with an error (exit-127): 2048 fifo, 8192 fifo.